Orthopedic AI & Agentic System

Complete Technical Guide: Datasets, Patient Records, Medical Imaging & Model Commands

System Overview: This document provides an exhaustive operational reference for the Orthopedic AI Chatbot, Multi-Agent Clinical Triage System, Fine-Tuning pipelines, synthetic and real-world orthopedic patient records, and medical scan datasets (X-Ray / MRI) with Kellgren-Lawrence (KL) grading.

1. Chatbot & LLM Fine-Tuning Datasets

The system combines high-grade orthopedic and clinical dialogue datasets into unified JSONL files formatted for instruction tuning (Qwen-2.5 / LoRA).

Dataset Source	Type & Scope	Records Filtered	License / Access
`ruslanmv/ai-medical-chatbot`	Multi-specialty patient-doctor dialogues	20,000 QA pairs	Open / HuggingFace
`jtatman/medical-sci-instruct-1m`	Complex scientific clinical reasoning	15,000 instruct pairs	Open / HuggingFace
`lavita/chatdoctor-*`	Real doctor-patient conversations (HealthCareMagic, iCliniq)	Orthopedic keywords filtered	Research Open
`Synthetic Ortho SFT`	Targeted TKA, THA, ACL, spine recovery & rehab protocols	510 QA pairs	Internal (Project)
TOTAL MERGED	**Consolidated Train + Val Splits**	**35,510 total records**	**Ready in backend/data/exports/**

Commands to Fetch and Fine-Tune Datasets:

Task	Execution Command
Fetch external datasets & merge into JSONL	`python backend/data/fetch_external_datasets.py`
Export synthetic Qwen SFT train/val splits	`python backend/data/export_sft_splits.py`
Run Qwen-2.5 1.5B/7B QLoRA Fine-Tuning	`python backend/data/train_qwen_lora.py`
Test fine-tuned model inference checkpoint	`python backend/data/test_lora_inference.py`

2. Orthopedic Patient Records Database

The local SQLite database (backend/patients.sqlite3) contains 50 comprehensive orthopedic patient records structured for clinical multi-agent workflows.

Database Schema Architecture:

- **patients**: Demographics, surgery type (TKA, THA, ACL Reconstruction, Lumbar Discectomy, Rotator Cuff Repair, Ankle Fusion), surgery date, surgeon name, implant details.
- **patient_medications**: Prescriptions, dosages, schedules, indication, risk notes (e.g., Enoxaparin, Tramadol, Celecoxib).
- **patient_rehab_exercises**: Prescribed rehab protocols, sets, reps, frequency, restrictions.
- **patient_documents**: Clinical notes, discharge summaries, op notes, extraction status.
- **patient_scans**: Medical imaging metadata (scan type, body part, laterality, KL grade, radiologist impression, local scan file path).

Database Action	Command / API Call
Seed 50 Orthopedic Patients with Full Records	`python backend/seed_ortho_patients.py`
Inspect DB Tables & Schema	`python backend/inspect_db.py`
Query Patient via Python API	`from patient_database import get_patient; pt = get_patient('PT-1005')`
Retrieve Patient List & Filters	`from patient_database import list_patients; pts = list_patients()`

3. Medical Imaging & Scan Datasets (X-Ray / MRI)

To support upcoming Computer Vision & Vision-Language Agents (VLM) for orthopedic assessments, the pipeline includes scan integration and automated linkage to patient records.

Dataset Name	Modality & Anatomy	Clinical Labels / Features	Access Method
`Kaggle: Knee Osteoarthritis`	X-Ray (AP views) Knee	Kellgren-Lawrence (KL) Grade 0 to 4 (9,700+ images)	Free / Kaggle API
`HuggingFace: medical-imaging`	Multimodal X-Ray / CT	Pathology classifications, fractures	Free / HuggingFace API
`Stanford MURA (Bone X-rays)`	X-Ray (Upper / Lower Extremities)	Musculoskeletal abnormalities, 40,000+ scans	Free Stanford Research DUA
`fastMRI (NYU / Meta)`	Knee & Spine MRI	Raw k-space & DICOM clinical knee MRI	fastmri.med.nyu.edu
`OAI (Osteoarthritis Initiative)`	Serial Knee Radiographs & 3T MRI	Gold standard longitudinal joint health study	NDA / NIMH Data Archive

Imaging Pipeline Commands:

Imaging Task	Execution Command
Link scan records to 50 patients (metadata + paths)	`python backend/download_scan_images.py --skip-download`
Download full Kaggle Knee X-Ray dataset	`python backend/download_scan_images.py --source kaggle`
Download HuggingFace medical images	`python backend/download_scan_images.py --source huggingface`
Download sample preview images directly	`python backend/download_scan_images.py --samples 50`

4. Real-World Hospital Clinical Data Repositories

For formal validation where simulated hospital data is insufficient, the following authenticated repositories provide real patient records:

- **ACORN Joint Replacement Registry**: Real de-identified Australian arthroplasty outcomes (TKA/THA) with PROMs (Oxford Knee Score). Creative Commons 4.0 license at *doi.org/10.26190/5ee17b8708b7e*.
- **PhysioNet MIMIC-IV**: 300,000+ hospital admissions including orthopedic surgical cases, real lab telemetry, medications, and discharge summaries. Access requires completing free CITI 'Data or Specimens Only Research' training at *physionet.org*.
- **Cancer Imaging Archive (TCIA)**: De-identified sarcoma and musculoskeletal CT/MRI scans in standard DICOM format.

5. Architecture & Git Collaboration Workflow

To ensure safe integration with your teammate's code before merging to main:

Step	Branch & Git Action	Description
1. Feature Branch	`git checkout rishp118-complete-medication-agent`	All datasets, DB scripts, scan loaders, and PDF docs committed here.
2. Push to Remote	`git push origin rishp118-complete-medication-agent`	Pushes branch cleanly to GitHub remote.
3. Merge with Teammate	`git fetch origin main` `git merge origin/main`	Pull your teammate's latest changes from main, resolve conflicts if any.
4. Pull Request to Main	`Create PR via GitHub / merge branch into main`	Safely merges medication agent, datasets, and patient DB into main project.

Generated for Orthopedic AI Assistant Project • All scripts and schemas reside under the backend/ directory.